United International University (UIU)

Dept. of Computer Science and Engineering (CSE)

Final Exam Year: **2025** Trimester: **Spring**

Course: **CSE 323/3711** Title: **Computer Networks**

Marks: **40** Time: **2 Hours**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

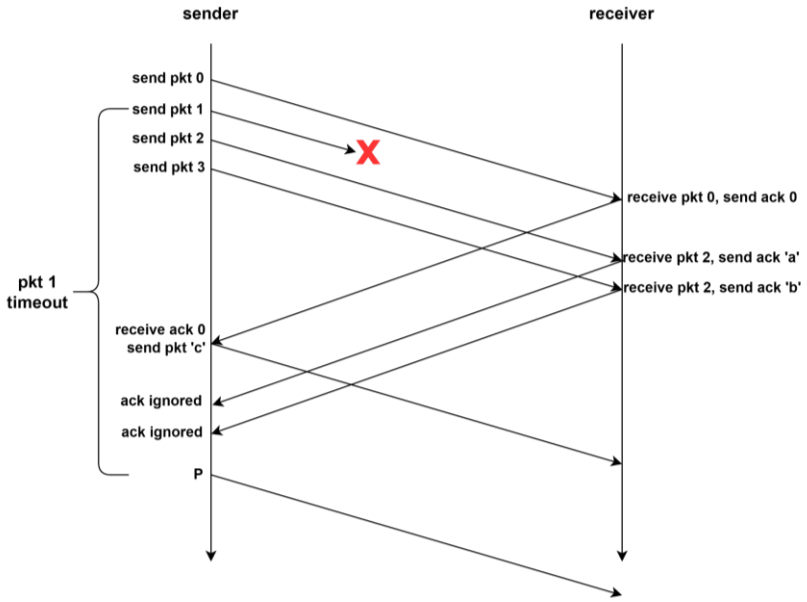
There are **4 (four)** questions. Answer **all** of them. All questions are of values indicated on the right-hand margin.

Q1. a) **Host A**, with **IP address 192.168.1.2** and **source port 5000**, sends an **HTTP request** to **Host B**, which has **IP address 192.168.1.10** and is running a web server listening on **port 80**. **Describe** how **multiplexing** and **demultiplexing** occur at the Transport layer. **[3]**

b) **Explain** the difference between **TCP** and **UDP** with **example scenarios** best suited for each of the protocols. **[2]**

c) Consider the **following scenario**, where the sender is scheduled to **send 10 packets** (Packets 0-9):

[2 + 1.5 + 1.5 = 5]



i) **Find** the values of a, b and c.

ii) After the **timeout**, what will be the **next sequence of packets** to be sent for **Go-Back-N** protocol?

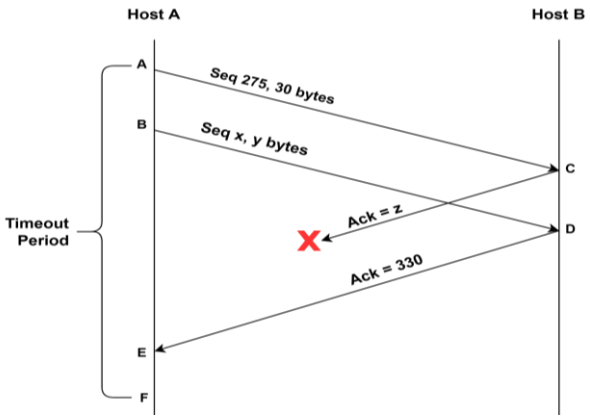
iii) After timeout, what will be the **next 4 sequence of packets** to be sent for **Selective Repeat** protocol?

Q2. a) Consider the following **TCP data transfe**r diagram: **[2 + 1.5 + 1.5 = 5]**

i) **Find** the values of **x, y, z**.

ii) Suppose, **Host A** still needs to send **50 bytes** of data; **explain** at **which point (E or F)** it will send the data. What will be the **sequence number** from that point?

iii) **Explain** what happens when the **Ack 330** arrives after the Timeout Period (with **necessary updates** to the diagram). What will be the **sequence number** of the next sent data?



b) Suppose that **two hosts** communicate over a **TCP connection**. **Explain** what happens if **network layer** delivers data **faster** than **application layer** removes data from **socket buffers**? **[2]**

c) Suppose that you are a **newly connected host** in a certain LAN and the **IP address** of the **DHCP server** is **143.161.10.31**. **Explain** the **4 (four) steps** of the DHCP process for allocating IP address for your host with a **diagram**. **[3]**

Q3. a) **Your host machine**, which is behind a **NAT router**, is running **CSGO game** on host **port # 3301** and **FaceTime video chat** on **port # 3302** with a **private IP address 10.0.0.1**. The **public IP address** of the **NAT router** is **138.76.29.7** that uses **port numbers in [5001, 5999]** for the NAT mechanism. Your friend **Arthur's host machine** (with **private IP address 10.0.0.2**), behind a **different NAT router**, is just running **FaceTime on port # 4501**. The **public IP address** of this NAT router is **137.91.12.44** that uses **port numbers from [7001, 7999]** for the NAT mechanism. You and Arthur are communicating via **FaceTime**. You are **sending a packet** (denoted by **PKT**) to Arthur. Now, **use the following diagram** addressing the following **2 queries**: **[4]**



- How will your **NAT router modify the PKT**?
- How will **Arthur's NAT router interpret and modify**, and ultimately, will his host is going to see it?

Show the ordered steps and **source IP, source port, destination IP, and destination port** for the packet for **each of these steps**.

b) An IP router with a **Maximum Transmission Unit (MTU)** of **1500 bytes** has received an IP packet of size **4480 bytes** with an **IP header** of length **20 bytes**. **How** will this packet be **fragmented**? **Show** each of the fragmented packets' **length, fragflag, and offset fields**. **[4]**

c) What **advantages** does **IPv6** have over **IPv4**? Write the following IPv6 in **compressed format**. **[2]**

2001:0db8:0000:1111:0000:0000:0000:0200

Q4. a) Given graph **G = (N, E)** on the right, where N is the set of routers and E is the set of links. Using **Dijkstra's link-state** routing algorithm, compute the **least cost path** from node U to all other nodes and **show** the resulting **forwarding table** for U, or the **least-cost-path tree** from U. **[5]**

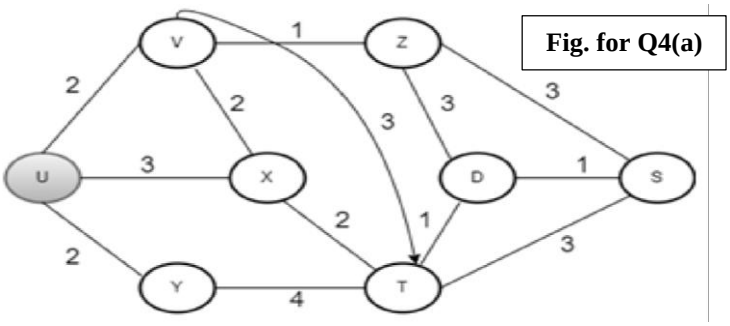
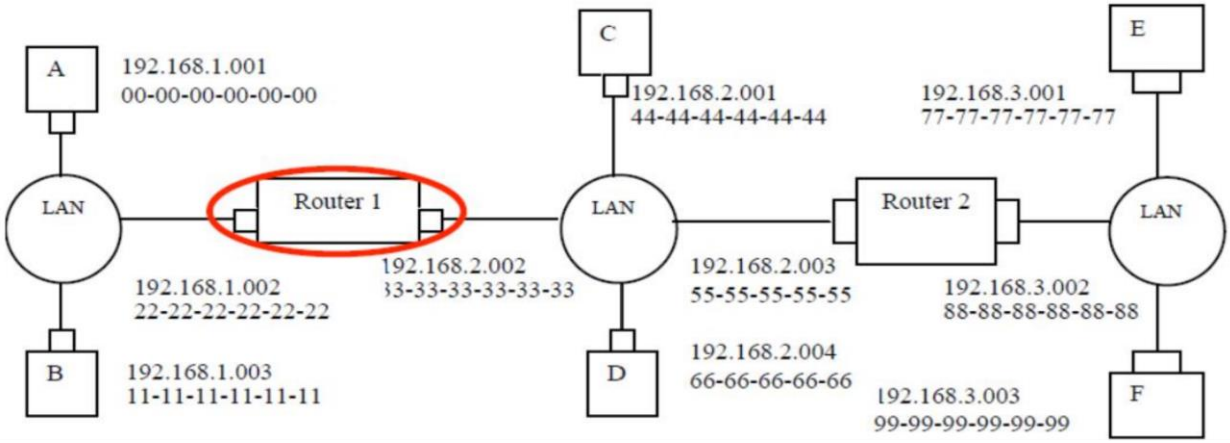


Fig. for Q4(a)

b) Suppose, **Host A** in the following **network** would like to send an **IP datagram** to **Host F**, and assume that **A's ARP table** is initially **empty**.



i. What will be the **destination MAC** of the **ARP query** sent by **Host A**? **[1]**
ii. **Which nodes** in the network will receive the **ARP query from Host A**? **[1]**
iii. **Which IP address** will respond to the **ARP query sent by Host A**? **[1]**
iv. In the **Ethernet Data frame** (containing the IP datagram destined to F) that is delivered to **Router 1**, what are the **source and destination IP and MAC addresses**? **[2]**